

Ibn Tofail University

Rigid body mechanics

Problem Set V

Exercise 1:

1. State Koenig's theorem relating to the angular momentum of a perfect rigid body S , in motion relative to a reference frame T_1 .
2. Prove this theorem.

Correction

1. Statement of Koenig's theorem:

For a rigid body S with center of mass G , the angular momentum about any point O is given by:

$$\vec{\sigma}_O(S/T_1) = \vec{\sigma}_G(S/T_1) + m\vec{OG} \times \vec{V}_G(S/T_1)$$

Where: - $\vec{\sigma}_O(S/T_1)$ is the angular momentum of S about point O relative to frame T_1 - $\vec{\sigma}_G(S/T_1)$ is the angular momentum of S about its center of mass G relative to frame T_1 - m is the total mass of the rigid body - $\vec{V}_G(S/T_1)$ is the velocity of the center of mass relative to frame T_1

2. Proof:

Let P be an arbitrary point of the rigid body S with mass dm . The angular momentum about O is:

$$\vec{\sigma}_O(S/T_1) = \int_{P \in S} \vec{OP} \times \vec{V}_P(S/T_1) dm$$

We can write $\vec{OP} = \vec{OG} + \vec{GP}$, so:

$$\begin{aligned} \vec{\sigma}_O(S/T_1) &= \int_{P \in S} (\vec{OG} + \vec{GP}) \times \vec{V}_P(S/T_1) dm \\ &= \vec{OG} \times \int_{P \in S} \vec{V}_P(S/T_1) dm + \int_{P \in S} \vec{GP} \times \vec{V}_P(S/T_1) dm \end{aligned}$$

For a rigid body, we have the velocity distribution formula:

$$\vec{V}_P(S/T_1) = \vec{V}_G(S/T_1) + \vec{\omega}_{S/T_1} \times \vec{GP}$$

Substituting this into the first integral:

$$\int_{P \in S} \vec{V}_P(S/T_1) dm = \int_{P \in S} [\vec{V}_G(S/T_1) + \vec{\omega}_{S/T_1} \times \vec{GP}] dm$$

$$= m\vec{V}_G(S/T_1) + \vec{\omega}_{S/T_1} \times \int_{P \in S} \vec{G}P dm$$

But $\int_{P \in S} \vec{G}P dm = 0$ by definition of the center of mass, so:

$$\int_{P \in S} \vec{V}_P(S/T_1) dm = m\vec{V}_G(S/T_1)$$

Now for the second integral:

$$\begin{aligned} \int_{P \in S} \vec{G}P \times \vec{V}_P(S/T_1) dm &= \int_{P \in S} \vec{G}P \times [\vec{V}_G(S/T_1) + \vec{\omega}_{S/T_1} \times \vec{G}P] dm \\ &= \int_{P \in S} \vec{G}P \times \vec{V}_G(S/T_1) dm + \int_{P \in S} \vec{G}P \times (\vec{\omega}_{S/T_1} \times \vec{G}P) dm \end{aligned}$$

The first term is:

$$\int_{P \in S} \vec{G}P \times \vec{V}_G(S/T_1) dm = \left(\int_{P \in S} \vec{G}P dm \right) \times \vec{V}_G(S/T_1) = 0$$

The second term is exactly the definition of the angular momentum about G :

$$\int_{P \in S} \vec{G}P \times (\vec{\omega}_{S/T_1} \times \vec{G}P) dm = \vec{\sigma}_G(S/T_1)$$

Therefore:

$$\vec{\sigma}_O(S/T_1) = \vec{OG} \times m\vec{V}_G(S/T_1) + \vec{\sigma}_G(S/T_1)$$

This completes the proof of Koenig's theorem.

Exercise 2:

A system S consists of a homogeneous rod S_1 of mass m_1 and length l , in motion, in the plane $O_1X_1Y_1$, about the axis O_1Z of the frame $T_1(O_1, X_1, Y_1, Z_1)$, and a disk S_2 , of mass m_2 , radius R , and center C , in motion about the axis CZ_1 (fig 1). We denote by θ the angle that indicates the position of the rod with respect to T_1 and by φ the angle that indicates the position of the disk with respect to T_1 . $(\vec{i}, \vec{j}, \vec{k}_1)$ is the basis of the frame $T(O, X, Y, Z_1)$ attached to the rod.

1. Determine the position of the center of mass G of the system.
2. Calculate the kinetic elements of system S with respect to T_1 , at point O_1 .

Correction

1. Position of the center of mass G :

Let's place the coordinate system such that O_1 is at the origin. The rod extends from O_1 to a point at distance l along the rod's axis.

For the homogeneous rod S_1 , its center of mass is at its midpoint:

$$G_1 : O_1 \vec{G}_1 = \frac{l}{2} \vec{i}$$

For the disk S_2 , its center C is at the end of the rod:

$$O_1 \vec{C} = l \vec{i}$$

The center of mass G of the entire system is given by:

$$O_1 \vec{G} = \frac{m_1 O_1 \vec{G}_1 + m_2 O_1 \vec{C}}{m_1 + m_2} = \frac{m_1 \cdot \frac{l}{2} \vec{i} + m_2 \cdot l \vec{i}}{m_1 + m_2} = \frac{l(m_1/2 + m_2)}{m_1 + m_2} \vec{i}$$

2. Kinetic elements at point O_1 :

The kinetic torsor has two elements: the momentum (resultant) and the angular momentum (moment).

a) Momentum:

$$\vec{P}(S/T_1) = (m_1 + m_2) \vec{V}_G(S/T_1)$$

We need $\vec{V}_G(S/T_1)$. From the position:

$$O_1 \vec{G} = \frac{l(m_1/2 + m_2)}{m_1 + m_2} \vec{i}$$

Differentiating with respect to time in frame T_1 :

$$\vec{V}_G(S/T_1) = \frac{d}{dt}(O_1 \vec{G})_{T_1} = \frac{l(m_1/2 + m_2)}{m_1 + m_2} \frac{d\vec{i}}{dt}$$

Since $\vec{i}$ is fixed in the rod frame which rotates with angular velocity $\dot{\theta} \vec{k}_1$ relative to T_1 :

$$\frac{d\vec{i}}{dt} = \vec{\omega}_{T/T_1} \times \vec{i} = \dot{\theta} \vec{k}_1 \times \vec{i} = \dot{\theta} \vec{j}$$

Therefore:

$$\vec{V}_G(S/T_1) = \frac{l(m_1/2 + m_2)}{m_1 + m_2} \dot{\theta} \vec{j}$$

And the momentum is:

$$\vec{P}(S/T_1) = (m_1 + m_2) \vec{V}_G(S/T_1) = l(m_1/2 + m_2) \dot{\theta} \vec{j}$$

b) Angular momentum at O_1 :

We use the additive property for composite systems:

$$\vec{\sigma}_{O_1}(S/T_1) = \vec{\sigma}_{O_1}(S_1/T_1) + \vec{\sigma}_{O_1}(S_2/T_1)$$

For the rod S_1 rotating about $O_1 Z$:

$$\vec{\sigma}_{O_1}(S_1/T_1) = I_{O_1 Z}(S_1) \dot{\theta} \vec{k}_1$$

Where $I_{O_1Z}(S_1)$ is the moment of inertia of the rod about axis O_1Z . For a thin rod of length l and mass m_1 rotating about one end:

$$I_{O_1Z}(S_1) = \frac{1}{3}m_1l^2$$

So:

$$\vec{\sigma}_{O_1}(S_1/T_1) = \frac{1}{3}m_1l^2\dot{\theta}\vec{k}_1$$

For the disk S_2 , we use Koenig's theorem:

$$\vec{\sigma}_{O_1}(S_2/T_1) = \vec{\sigma}_C(S_2/T_1) + m_2\vec{O_1C} \times \vec{V}_C(S_2/T_1)$$

The disk rotates about CZ_1 with angular velocity $\dot{\varphi}\vec{k}_1$, so:

$$\vec{\sigma}_C(S_2/T_1) = I_{CZ}(S_2)\dot{\varphi}\vec{k}_1$$

For a disk about its central axis perpendicular to its plane:

$$I_{CZ}(S_2) = \frac{1}{2}m_2R^2$$

Now, $\vec{O_1C} = l\vec{i}$ and $\vec{V}_C(S_2/T_1) = \frac{d}{dt}(l\vec{i})_{T_1} = l\dot{\theta}\vec{j}$

So:

$$m_2\vec{O_1C} \times \vec{V}_C(S_2/T_1) = m_2(l\vec{i}) \times (l\dot{\theta}\vec{j}) = m_2l^2\dot{\theta}(\vec{i} \times \vec{j}) = m_2l^2\dot{\theta}\vec{k}_1$$

Therefore:

$$\vec{\sigma}_{O_1}(S_2/T_1) = \frac{1}{2}m_2R^2\dot{\varphi}\vec{k}_1 + m_2l^2\dot{\theta}\vec{k}_1$$

The total angular momentum is:

$$\begin{aligned}\vec{\sigma}_{O_1}(S/T_1) &= \left(\frac{1}{3}m_1l^2\dot{\theta} + m_2l^2\dot{\theta} + \frac{1}{2}m_2R^2\dot{\varphi} \right) \vec{k}_1 \\ &= \left(l^2 \left(\frac{m_1}{3} + m_2 \right) \dot{\theta} + \frac{1}{2}m_2R^2\dot{\varphi} \right) \vec{k}_1\end{aligned}$$

The kinetic torsor at O_1 is therefore:

$$\tau(S/T_1) = \left[l(m_1/2 + m_2)\dot{\theta}\vec{j}, \left(l^2 \left(\frac{m_1}{3} + m_2 \right) \dot{\theta} + \frac{1}{2}m_2R^2\dot{\varphi} \right) \vec{k}_1 \right]$$

Exercise 3:

A homogeneous rod of length l and mass m , rotates, in the vertical plane XOY , about the axis OZ of a reference frame $R(O, X, Y, Z)$ assumed to be Galilean. The position of the rod with respect to R is indicated by the angle θ (fig 2). We admit that the torsor of contact actions at O exerted on the rod reduces to the reaction $\vec{R}$.

1. By applying the angular momentum theorem to the rod, at O , find a first integral of motion, knowing that $\dot{\theta} = \omega_0$ when $\theta = 0$.

2. By applying the momentum theorem, determine the expression of the reaction.

Correction

1. First integral of motion using angular momentum theorem:

The angular momentum theorem at point O states:

$$\frac{d\vec{\sigma}_O}{dt} = \vec{M}_O(\text{external forces})$$

The external forces are: - Weight: mg acting at the center of mass G - Reaction at O : $\vec{R}$

The moment of the reaction at O about O is zero. The moment of the weight about O is:

$$\vec{M}_O(\text{weight}) = \vec{OG} \times m\vec{g}$$

The center of mass G is at distance $l/2$ from O along the rod. In the rotating frame attached to the rod:

$$\vec{OG} = \frac{l}{2}\vec{i}$$

Where $\vec{i}$ is the unit vector along the rod. The gravitational acceleration is:

$$\vec{g} = -g\vec{j}$$

The angular momentum of the rod about O is:

$$\vec{\sigma}_O = I_O\dot{\theta}\vec{k}$$

Where I_O is the moment of inertia of the rod about O . For a thin rod of length l and mass m rotating about one end:

$$I_O = \frac{1}{3}ml^2$$

So:

$$\vec{\sigma}_O = \frac{1}{3}ml^2\dot{\theta}\vec{k}$$

Now, the angular momentum theorem gives:

$$\frac{d}{dt} \left(\frac{1}{3}ml^2\dot{\theta}\vec{k} \right) = \vec{OG} \times m\vec{g}$$

The left side is:

$$\frac{1}{3}ml^2\ddot{\theta}\vec{k}$$

The right side is:

$$\vec{OG} \times m\vec{g} = \left(\frac{l}{2}\vec{i} \right) \times (-mg\vec{j}) = -\frac{mgl}{2}(\vec{i} \times \vec{j}) = -\frac{mgl}{2}\vec{k}$$

So we have:

$$\frac{1}{3}ml^2\ddot{\theta} = -\frac{mgl}{2}$$

Simplifying:

$$\ddot{\theta} = -\frac{3g}{2l}$$

This is a differential equation we can integrate. Multiply both sides by $2\dot{\theta}$:

$$2\dot{\theta}\ddot{\theta} = -\frac{3g}{l}\dot{\theta}$$

Integrate with respect to time:

$$\dot{\theta}^2 = -\frac{3g}{l}\theta + C$$

Using the initial condition $\dot{\theta} = \omega_0$ when $\theta = 0$:

$$\omega_0^2 = C$$

So the first integral is:

$$\dot{\theta}^2 = \omega_0^2 - \frac{3g}{l}\theta$$

2. Reaction using momentum theorem:

The momentum theorem states:

$$m\vec{a}_G = \vec{R} + m\vec{g}$$

We need the acceleration of the center of mass G . The position of G is:

$$\vec{OG} = \frac{l}{2}\vec{i}$$

Differentiating twice with respect to time:

$$\vec{V}_G = \frac{l}{2}\dot{\theta}\vec{j}$$

$$\vec{a}_G = \frac{l}{2}\ddot{\theta}\vec{j} - \frac{l}{2}\dot{\theta}^2\vec{i}$$

We already have $\ddot{\theta} = -\frac{3g}{2l}$ and $\dot{\theta}^2 = \omega_0^2 - \frac{3g}{l}\theta$

So:

$$\begin{aligned}\vec{a}_G &= \frac{l}{2}\left(-\frac{3g}{2l}\right)\vec{j} - \frac{l}{2}\left(\omega_0^2 - \frac{3g}{l}\theta\right)\vec{i} \\ &= -\frac{3g}{4}\vec{j} - \frac{l}{2}\left(\omega_0^2 - \frac{3g}{l}\theta\right)\vec{i}\end{aligned}$$

Now from the momentum theorem:

$$\vec{R} = m\vec{a}_G - m\vec{g} = m\vec{a}_G + mg\vec{j}$$

Substituting $\vec{a}_G$:

$$\vec{R} = m\left[-\frac{l}{2}\left(\omega_0^2 - \frac{3g}{l}\theta\right)\vec{i} - \frac{3g}{4}\vec{j}\right] + mg\vec{j}$$

$$\begin{aligned}
&= -\frac{ml}{2} \left(\omega_0^2 - \frac{3g}{l}\theta \right) \vec{i} + \left(mg - \frac{3mg}{4} \right) \vec{j} \\
&= -\frac{ml}{2} \left(\omega_0^2 - \frac{3g}{l}\theta \right) \vec{i} + \frac{mg}{4} \vec{j}
\end{aligned}$$

So the reaction at O is:

$$\vec{R} = -\frac{ml}{2} \left(\omega_0^2 - \frac{3g}{l}\theta \right) \vec{i} + \frac{mg}{4} \vec{j}$$

Exercise 4:

A homogeneous bar AB , of length $2l$ and mass m , rests on a point E located at a height h above the horizontal plane as shown in figure 3. Under the action of the force $\vec{F}_A$, point A moves on the axis OX with a constant velocity v_A . The contacts at points A and E are frictionless. The position of the bar, with respect to the terrestrial reference frame $R(O, X, Y, Z)$ assumed to be Galilean, is defined by the angle θ . The distance OH is constant.

1. Express $\dot{\theta}$ and $\ddot{\theta}$ as a function of θ and v_A .
2. Determine, in the basis of R and as a function of $\dot{\theta}$, the momentum of bar AB and its angular momentum at point A .
3. By applying the angular momentum theorem to bar AB at point A , determine the expression of the reaction $\vec{R}_E$ exerted on the bar at point E .

Correction

1. Expressing $\dot{\theta}$ and $\ddot{\theta}$:

From the geometry, point A has coordinates $(x_A, 0)$ and point E has coordinates $(0, h)$. The bar has length $2l$.

The distance between A and E is:

$$AE = \sqrt{x_A^2 + h^2}$$

But also from the bar's geometry, the vertical projection gives:

$$h = 2l \sin \theta$$

And the horizontal distance from A to the vertical through E is:

$$x_A = 2l \cos \theta - OH$$

Since OH is constant and $v_A = \dot{x}_A$ is constant, we have:

$$x_A = 2l \cos \theta - OH$$

$$\dot{x}_A = -2l\dot{\theta} \sin \theta = v_A$$

Therefore:

$$\dot{\theta} = -\frac{v_A}{2l \sin \theta}$$

Differentiating again:

$$\ddot{\theta} = \frac{v_A}{2l} \cdot \frac{\dot{\theta} \cos \theta}{\sin^2 \theta} = \frac{v_A}{2l} \cdot \left(-\frac{v_A}{2l \sin \theta} \right) \frac{\cos \theta}{\sin^2 \theta} = -\frac{v_A^2 \cos \theta}{4l^2 \sin^3 \theta}$$

2. Momentum and angular momentum at point A:

The center of mass G is at the midpoint of the bar. Its position relative to A is:

$$\vec{AG} = l(\cos \theta \vec{i} + \sin \theta \vec{j})$$

The velocity of G is:

$$\vec{V}_G = \vec{V}_A + \vec{\omega} \times \vec{AG}$$

Since $\vec{V}_A = v_A \vec{i}$ and $\vec{\omega} = \dot{\theta} \vec{k}$:

$$\begin{aligned} \vec{V}_G &= v_A \vec{i} + \dot{\theta} \vec{k} \times l(\cos \theta \vec{i} + \sin \theta \vec{j}) \\ &= v_A \vec{i} + l\dot{\theta}(-\sin \theta \vec{i} + \cos \theta \vec{j}) \\ &= (v_A - l\dot{\theta} \sin \theta) \vec{i} + l\dot{\theta} \cos \theta \vec{j} \end{aligned}$$

The momentum is:

$$\vec{P} = m\vec{V}_G = m(v_A - l\dot{\theta} \sin \theta) \vec{i} + ml\dot{\theta} \cos \theta \vec{j}$$

The angular momentum at point A is:

$$\vec{\sigma}_A = I_A \dot{\theta} \vec{k}$$

Where I_A is the moment of inertia of the bar about point A . For a thin bar of length $2l$ and mass m about one end:

$$I_A = \frac{1}{3}m(2l)^2 = \frac{4}{3}ml^2$$

So:

$$\vec{\sigma}_A = \frac{4}{3}ml^2 \dot{\theta} \vec{k}$$

3. Reaction $\vec{R}_E$ using angular momentum theorem at A:

The angular momentum theorem at point A states:

$$\frac{d\vec{\sigma}_A}{dt} = \vec{M}_A(\text{external forces})$$

The external forces are: - Weight: mg acting at G - Reaction at E : $\vec{R}_E = R_{Ex} \vec{i} + R_{Ey} \vec{j}$ - Force at A : $\vec{F}_A$ (but its moment about A is zero)

The moment of the weight about A is:

$$\vec{M}_A(\text{weight}) = \vec{AG} \times m\vec{g} = l(\cos \theta \vec{i} + \sin \theta \vec{j}) \times (-mg\vec{j}) = -mgl \cos \theta (\vec{i} \times \vec{j}) = -mgl \cos \theta \vec{k}$$

The moment of the reaction at E about A is:

$$\vec{M}_A(\vec{R}_E) = \vec{A}E \times \vec{R}_E$$

We have $\vec{A}E = -x_A \vec{i} + h \vec{j} = (-2l \cos \theta + OH) \vec{i} + 2l \sin \theta \vec{j}$
So:

$$\vec{A}E \times \vec{R}_E = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2l \cos \theta + OH & 2l \sin \theta & 0 \\ R_{Ex} & R_{Ey} & 0 \end{vmatrix} = [(-2l \cos \theta + OH)R_{Ey} - (2l \sin \theta)R_{Ex}] \vec{k}$$

The left side of the angular momentum theorem is:

$$\frac{d\vec{\sigma}_A}{dt} = \frac{4}{3}ml^2\ddot{\theta}\vec{k}$$

So the angular momentum theorem gives:

$$\frac{4}{3}ml^2\ddot{\theta} = -mgl \cos \theta + [(-2l \cos \theta + OH)R_{Ey} - (2l \sin \theta)R_{Ex}]$$

We need another equation to find both components of $\vec{R}_E$. Let's use the momentum theorem:

$$m\vec{a}_G = \vec{F}_A + \vec{R}_E + m\vec{g}$$

The acceleration of G is:

$$\begin{aligned} \vec{a}_G &= \frac{d\vec{V}_G}{dt} = \frac{d}{dt}[(v_A - l\dot{\theta} \sin \theta)\vec{i} + l\dot{\theta} \cos \theta \vec{j}] \\ &= [-l\ddot{\theta} \sin \theta - l\dot{\theta}^2 \cos \theta]\vec{i} + [l\ddot{\theta} \cos \theta - l\dot{\theta}^2 \sin \theta]\vec{j} \end{aligned}$$

So the momentum theorem gives:

$$m[-l\ddot{\theta} \sin \theta - l\dot{\theta}^2 \cos \theta] = F_{Ax} + R_{Ex}$$

$$m[l\ddot{\theta} \cos \theta - l\dot{\theta}^2 \sin \theta] = R_{Ey} - mg$$

We can solve these equations to find R_{Ex} and R_{Ey} , but the expression will be quite complex in terms of θ , v_A , l , and OH .

The final expression for $\vec{R}_E$ is:

$$\vec{R}_E = \left(-m[-l\ddot{\theta} \sin \theta - l\dot{\theta}^2 \cos \theta] - F_{Ax}\right) \vec{i} + \left(m[l\ddot{\theta} \cos \theta - l\dot{\theta}^2 \sin \theta] + mg\right) \vec{j}$$

Where $\dot{\theta}$ and $\ddot{\theta}$ are given in part 1.

Exercise 5:

A material point A of mass m is fixed on the perimeter of a hoop D , of center C , radius R , and mass M . This system S rolls without slipping on the axis OX of a terrestrial reference frame $R(O, X, Y, Z)$ assumed to be Galilean. The motion takes place in the

plane XOY . We denote by θ the position of point A with respect to the hoop D , and I the geometric contact point (fig 4).

1. What is the number of degrees of freedom of the system?
2. Calculate the velocities of points I , C , and A with respect to R .
3. Find the momentum of S .
4. Determine the differential equation of motion, in θ , of S by applying the angular momentum theorem at point I .
5. What is the nature of the motion of S ?

Correction

1. Number of degrees of freedom:

The system consists of a hoop with a fixed point on its perimeter. The hoop rolls without slipping on a fixed line. This imposes one constraint. The system has one degree of freedom, which can be described by the angle θ .

2. Velocities of points I , C , and A :

Let ϕ be the rotation angle of the hoop about its center C . Due to rolling without slipping, when the hoop rotates by an angle ϕ , its center moves by $R\phi$ along OX . So the position of center C is:

$$\vec{OC} = R\phi\vec{i} + R\vec{j}$$

The velocity of C is:

$$\vec{V}_C = R\dot{\phi}\vec{i}$$

Point I is the contact point. Its position is:

$$\vec{OI} = R\phi\vec{i}$$

Since the hoop rolls without slipping, point I has zero velocity relative to the ground:

$$\vec{V}_I = \vec{0}$$

Point A is fixed on the hoop at angular position θ from the vertical. Its position relative to C is:

$$\vec{CA} = R\sin\theta\vec{i} - R\cos\theta\vec{j}$$

So the position of A is:

$$\vec{OA} = \vec{OC} + \vec{CA} = (R\phi + R\sin\theta)\vec{i} + (R - R\cos\theta)\vec{j}$$

The velocity of A is:

$$\begin{aligned}\vec{V}_A &= \frac{d\vec{OA}}{dt} = R\dot{\phi}\vec{i} + R\dot{\theta}\cos\theta\vec{i} + R\dot{\theta}\sin\theta\vec{j} \\ &= R(\dot{\phi} + \dot{\theta}\cos\theta)\vec{i} + R\dot{\theta}\sin\theta\vec{j}\end{aligned}$$

3. Momentum of S :

The system S consists of the hoop D (mass M) and the point mass A (mass m).

The momentum is:

$$\begin{aligned}\vec{P}(S/R) &= M\vec{V}_C + m\vec{V}_A \\ &= MR\dot{\phi}\vec{i} + m[R(\dot{\phi} + \dot{\theta}\cos\theta)\vec{i} + R\dot{\theta}\sin\theta\vec{j}] \\ &= R[(M+m)\dot{\phi} + m\dot{\theta}\cos\theta]\vec{i} + mR\dot{\theta}\sin\theta\vec{j}\end{aligned}$$

4. Differential equation of motion using angular momentum theorem at I :

The angular momentum theorem at point I states:

$$\frac{d\vec{\sigma}_I}{dt} = \vec{M}_I(\text{external forces})$$

The external forces are: - Weight of hoop: Mg acting at C - Weight of point mass: mg acting at A - Normal reaction at I (but its moment about I is zero) - Friction at I (but its moment about I is zero)

The angular momentum about I has contributions from both the hoop and the point mass.

For the hoop, we use Koenig's theorem:

$$\vec{\sigma}_I(D/R) = \vec{\sigma}_C(D/R) + M\vec{IC} \times \vec{V}_C$$

We have $\vec{IC} = R\vec{j}$ and $\vec{V}_C = R\dot{\phi}\vec{i}$, so:

$$M\vec{IC} \times \vec{V}_C = M(R\vec{j}) \times (R\dot{\phi}\vec{i}) = MR^2\dot{\phi}(\vec{j} \times \vec{i}) = -MR^2\dot{\phi}\vec{k}$$

The hoop rotates about its center with angular velocity $\dot{\phi}$, so:

$$\vec{\sigma}_C(D/R) = I_C\dot{\phi}\vec{k}$$

For a hoop about its center:

$$I_C = MR^2$$

So:

$$\vec{\sigma}_C(D/R) = MR^2\dot{\phi}\vec{k}$$

Therefore:

$$\vec{\sigma}_I(D/R) = MR^2\dot{\phi}\vec{k} - MR^2\dot{\phi}\vec{k} = 0$$

This is interesting: the angular momentum of a rolling hoop about the contact point is zero.

For the point mass A :

$$\vec{\sigma}_I(A/R) = \vec{IA} \times m\vec{V}_A$$

We have $\vec{IA} = \vec{OA} - \vec{OI} = (R\phi + R\sin\theta - R\phi)\vec{i} + (R - R\cos\theta)\vec{j} = R\sin\theta\vec{i} + R(1 - \cos\theta)\vec{j}$

And $\vec{V}_A = R(\dot{\phi} + \dot{\theta}\cos\theta)\vec{i} + R\dot{\theta}\sin\theta\vec{j}$

So:

$$\vec{\sigma}_I(A/R) = m \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ R\sin\theta & R(1 - \cos\theta) & 0 \\ R(\dot{\phi} + \dot{\theta}\cos\theta) & R\dot{\theta}\sin\theta & 0 \end{vmatrix} = mR^2[\sin\theta\dot{\theta}\sin\theta - (1 - \cos\theta)(\dot{\phi} + \dot{\theta}\cos\theta)]\vec{k}$$

$$= mR^2[\dot{\theta} \sin^2 \theta - (1 - \cos \theta)\dot{\phi} - (1 - \cos \theta)\dot{\theta} \cos \theta]\vec{k}$$

The total angular momentum is:

$$\vec{\sigma}_I(S/R) = mR^2[\dot{\theta} \sin^2 \theta - (1 - \cos \theta)\dot{\phi} - (1 - \cos \theta)\dot{\theta} \cos \theta]\vec{k}$$

Now, the moment of external forces about I :

$$\begin{aligned}\vec{M}_I &= I\vec{C} \times M\vec{g} + I\vec{A} \times m\vec{g} \\ &= (R\vec{j}) \times (-Mg\vec{j}) + [R \sin \theta \vec{i} + R(1 - \cos \theta)\vec{j}] \times (-mg\vec{j}) \\ &= 0 + (-mgR \sin \theta)(\vec{i} \times \vec{j}) = -mgR \sin \theta \vec{k}\end{aligned}$$

The angular momentum theorem gives:

$$\frac{d}{dt} \left\{ mR^2[\dot{\theta} \sin^2 \theta - (1 - \cos \theta)\dot{\phi} - (1 - \cos \theta)\dot{\theta} \cos \theta] \right\} = -mgR \sin \theta$$

This is the differential equation of motion. It can be simplified using trigonometric identities.

5. Nature of the motion:

The motion is oscillatory about a stable equilibrium position. The system behaves like a pendulum, with the point mass A oscillating about the vertical position.